ISO 9001:2008 Certified Institute

JAVA INSTITUTE FOR ADVANCED TECHNOLOGY

Department of Examinations

Unit Name	Object Oriented Design Patterns II
Unit Id	JIAT/OODP II
Assessment Id	JIAT/OODP II /EX/01
Assessment Summary	This assessment requires the application of Object-Oriented Design Patterns which they studied at the class to solve practical, real-world problems. Students must analyze given scenarios, select appropriate Design Patterns, and provide a justified implementation. The assessment is structured into five tasks, each contributing equally to the final grade. All tasks are compulsory and must be submitted via the online portal by the specified deadline.
Duration	2 Weeks
Submission Via	Online (Student Portal)
Document Format	Document Format (Pdf)

GUIDELINES FOR CANDIDATES

- Your studies will be governed by the Java Institute Academic Regulations on Assessment, Progression and Awards.
- Students are expected to use reference books, the Internet, journals and other similar sources in order to accomplish the task specified above.
- Students are expected to refrain from repeating any content in their research document.
- At the reassessment attempt, the mark is capped and the maximum mark that can be achieved is 40%.

CHEATING AND PLAGIARISM

Both cheating and plagiarism are totally unacceptable and the Institute maintains a strict policy against them. It is YOUR responsibility to be aware of this policy and to act accordingly.

The basic principles are:

- Don't pass off anyone else's work as your own, including coding examples. This is plagiarism and is viewed extremely seriously by the Institute.
- Don't submit a piece of work in whole or in part that has already been submitted for assessment elsewhere. This is called duplication and, like plagiarism, is viewed extremely seriously by the Institute.
- Always acknowledge all of the sources that you have used in your assessment or project.
- If you are using the exact words of another person, always put them in quotation marks.
- Check that you know whether the assessment is to be produced individually or whether you can work with others.
- If you are doing group work, be sure about what you are supposed to do on your own.
- Never make up or falsify data to prove your point.
- Never allow others to copy your work.
- Never lend disks, memory sticks, or copies of your coursework to any other student. in the Institute. This may lead to you being accused of collusion.

Assessment Summary

This assessment evaluates your ability to apply object-oriented design patterns to solve real-world problems in a Smart City Management System (SmartCity360). Each task below requires you to analyze a problem scenario and response using exactly one appropriate design pattern, selected from the set provided. All questions are compulsory.

SmartCity360 – A Smart City Management Platform

SmartCity360 is a digital solution that manages city infrastructure, devices, services, and resident interactions. It integrates systems such as waste management, energy monitoring, sensor networks, and public administration.

As a senior developer at UrbanTech Solutions, your role is to architect maintainable, efficient, and secure components using design patterns.

In this assessment, you will work on five parts (A, B, C, D, and E), each focusing on different components of the SmartCity360 system and requiring you to apply appropriate object-oriented design patterns to solve real-world problems. Part F provides instructions for the final presentation and report.

Part A - Dynamic Security Wrappers for Data Streams

SmartCity360 must secure data streams from energy meters with layers such as encryption, authentication, and compression, which should be addable without altering the core logic.

1. Identify and justify the most suitable design pattern for this requirement. **(3 marks)**
2. Explain how the pattern supports flexible and layered enhancements to object functionality. **(4 marks)**
3. Evaluate the long-term benefits of this pattern in terms of reusability and system extensibility. **(8 marks)**

(Total = 15 marks)

Part B - Hierarchical Structure of Smart Zones

City zones (city → district → street → building) need to be represented hierarchically so that commands like “shutdown power” or “report usage” can be applied uniformly at any level.

1. Identify the most appropriate design pattern to model such a structure. **(3 marks)**
2. Provide a UML class diagram to demonstrate the selected design pattern’s mechanism. **(4 marks)**
3. Discuss its usefulness for managing a large and growing city infrastructure. **(8 marks)**

(Total = 15 marks)

Part C - Escalation of Citizen Complaints

When a resident submits a complaint, the system should escalate it from local staff to supervisors to city managers depending on severity.

1. Identify the most appropriate pattern to handle dynamic escalation paths. **(3 marks)**
2. Explain how this pattern enables decoupled request handling. **(4 marks)**
3. Evaluate its benefits and drawbacks in terms of flexibility and traceability. **(8 marks)**

(Total = 15 marks)

Part D - Efficient Management of Smart Street Lights

The city uses thousands of smart streetlights with identical behavior and configuration but different locations. The system must minimize memory usage.

1. Identify and justify the pattern that best optimizes resource use in this scenario. **(3 marks)**
2. Explain how this pattern handles shared (intrinsic) and unique (extrinsic) properties. **(4 marks)**
3. Analyze the performance impact when scaling to tens of thousands of similar objects. **(8 marks)**

(Total = 15 marks)

Part E - Interpreting City Policies Defined in DSL

The city's operational rules (e.g., noise limits, parking hours) are defined in a mini-language that must be parsed and evaluated dynamically by the system.

1. Identify the most appropriate design pattern for interpreting this rule language. **(3 marks)**
2. Provide a UML class diagram to demonstrate the selected design pattern's mechanism. **(4 marks)**
3. Critically assess the pattern's suitability for rule updates and extensibility. **(8 marks)**

(Total = 15 marks)

Part F – Final Project Demonstration, VIVA, and Documentation

1. You are expected to develop the system based on the given scenario, applying the appropriate object-oriented design patterns.

2. You will demonstrate your application during a VIVA session, where you will explain your use of design patterns, discuss architectural choices, and answer questions related to the design, functionality, and overall system structure. **(15 marks)**

3. You are expected to prepare comprehensive project documentation, including a UML class diagram and working code, to illustrate the design and functionality of your application. **(10 marks)**

4. You must submit a **10–15-minute recorded demonstration video** showcasing your implemented system, its key functionalities, and the application of the selected design patterns. The video should clearly demonstrate how your solution addresses the given scenario.

5. **Final Submission Requirements:**

    a. **Project Documentation (Project Document):** Upload the final project report in **PDF format** to the **Project Document** section of the Student Portal.

    b. **Project Source Code (Project File):** Upload the complete project source code, including all required files and resources, as a **ZIP file** to the **Project File** section of the Student Portal.

    c. **Demonstration Video:** Upload the **10–15-minute** recorded demonstration video in **MP4 format** to the **Demonstration Video** section of the Student Portal.

(Total = 25 marks)

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